



BLOCKCHAIN INDIA CHALLENGE

Blockchain-Enabled Public Distribution System (VikshitPDS)

Transparent • Accountable • Leak-Proof

Immutable Grain Procurement Ledger

Aadhaar-Linked Beneficiary Verification

Smart Contract-Driven Entitlement Release

AI-Powered Leakage Detection & Analytics

01 | Problem Statement

₹2.05L Cr

Annual food subsidy budget (FY25)

35–40%

Estimated leakage & diversion

81 Crore

Beneficiaries under NFSA

5.4 Lakh

Fair Price Shops nationwide

Ghost Beneficiaries

Millions of fake/duplicate ration cards systematically divert food grains to black markets. Manual de-duplication is slow and easily circumvented.

Mid-Chain Diversion

Grains are siphoned between warehouses and Fair Price Shops. No real-time tracking means diversions go undetected for months.

Opaque Records

Paper-based FPS registers are trivially falsified. No common source of truth exists across central, state, district and shop levels.

Exclusion Errors

Genuine beneficiaries are denied entitlements due to biometric failures, poor connectivity, or data mismatches — a critical last-mile failure.

GrainChain India

A Hyperledger Fabric-powered permissioned blockchain creating an immutable, shared record of every grain transaction — from MSP procurement to last-mile FPS delivery — for 81 crore beneficiaries.



Immutable Grain Ledger

Every procurement, movement & delivery anchored on-chain as a tamper-proof record.



Aadhaar Oracle Verification

Live biometric oracle confirms beneficiary identity before any entitlement is released.



Smart Entitlement Contracts

Chaincode auto-triggers quota release on eligibility match — zero manual intervention.



AI Leakage Analytics

ML engine flags anomalies in offtake, stock movement, and transaction velocity in real time.



Offline FPS Mobile dApp

Android app for FPS dealers syncs transactions when connectivity is restored.



IoT-GPS Blockchain Oracle

Vehicle GPS telemetry written directly to the chain — full cold-chain traceability.

Why Blockchain for PDS?

Immutability

FPS offtake entries recorded on-chain cannot be altered — fraudulent records become impossible.

Decentralization

Centre, state agencies, FCI, district offices and FPS shops all hold the same verified ledger copy.

Transparency

Every stakeholder sees the exact grain journey with role-based access — nothing hidden, nothing lost.

Smart Automation

Entitlements self-execute when eligibility is confirmed — removing the human manipulation layer entirely.

Stakeholder Trust Flow

FCI / NAFED

Procurement & central storage

Transport / Logistics

Tracked movement with IoT-GPS

State Food Dept.

Allocation & district monitoring

Fair Price Shop (FPS)

Last-mile distribution node

Beneficiary (NFSA)

Aadhaar-verified offtake

04 | Technical Approach & Architecture

Citizen & Dealer Interface

ONORC Web Portal | FPS Android dApp (offline-capable) | Beneficiary SMS/IVRS | Grievance Portal

API Gateway & Integration Middleware

REST / GraphQL | Aadhaar eKYC Bridge | PFMS & DBTL APIs | State NIC Integration | Event Bus

Smart Contract Layer (Chaincode – Go)

Procurement Chaincode | Allocation Chaincode | Entitlement Chaincode | Grievance & Audit Chaincode

Hyperledger Fabric 2.5 Network

Raft Orderer | FCI Peer | State Agency Peers | FPS Peers | CAG Audit Node | MSP / x.509 Certs

Data & Oracle Layer

IPFS (document store) | Aadhaar Oracle | IoT-GPS Oracle | NFSA DB sync | CouchDB world-state

05 | Benchmarking & Differentiation

Dimension	Current Digitised PDS	Existing State Pilots	VikshitPDS ✦
Record Integrity	Centralised DB — mutable	Partial digitisation	Immutable on-chain ledger
Beneficiary Auth	Aadhaar PoS — offline gaps	Online Aadhaar only	Aadhaar oracle + OTP fallback
Grain Tracking	Manual GRN paper trail	SMS-based updates	IoT-GPS + blockchain oracle
Leakage Detection	Post-audit only	None / ad-hoc	Real-time AI anomaly engine
Smart Contracts	None	None	Full lifecycle chaincode (Go)
Interoperability	Siloed state systems	State-level only	ONORC + national federation
Audit Trail	Manual, incomplete	Partial	Immutable CAG-ready ledger

✦ GrainChain India uniquely combines Hyperledger Fabric + AI anomaly detection + live IoT-GPS oracles — no existing PDS solution offers all three.

Platform: Hyperledger Fabric 2.5

Zero Cryptocurrency

No token or coin — completely compliant with RBI regulations and Indian financial laws.

MSP + x.509 Identity

Only credentialed entities (FCI, state agencies, FPS dealers) can join — no anonymous nodes.

Private Data Collections

Sensitive beneficiary data shared only with authorised parties — never exposed on full ledger.

Raft Consensus (CFT)

Crash-fault-tolerant ordering for government-grade uptime with no cryptocurrency incentives.

Go / Node.js Chaincode

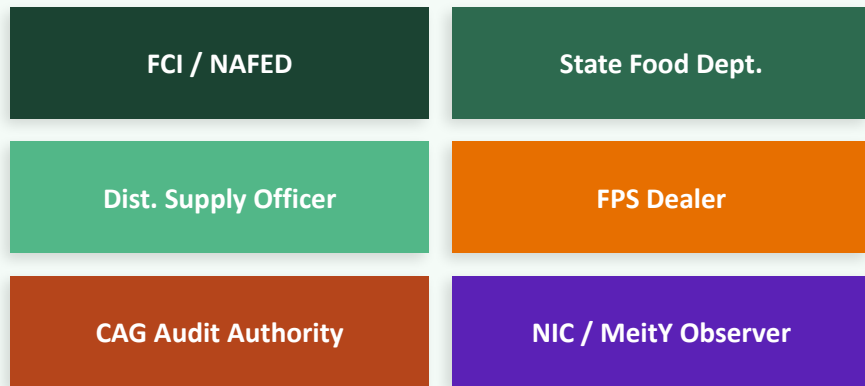
Strongly typed, auditable smart contracts reviewable by NIC and CERT-In teams.

Linux Foundation Backed

Can be licensed to both countries equivalent to India

07 | Permissioned Blockchain & Smart Contracts

Permissioned Network Nodes



All nodes hold identical ledger copies via Gossip Protocol. MSP enforces role-based identity and access.

Chaincode (Smart Contract) Workflows

Procurement Chaincode

Records FCI / state agency grain intake at MSP centres; validates weight, quality certs and warehouse receipts; emits a signed on-chain receipt.

Allocation Chaincode

Computes monthly FPS quota from NFSA entitlement rules; auto-generates signed allocation orders distributed to state and district peers.

Entitlement Chaincode

Queries Aadhaar oracle for biometric confirmation; atomically releases grain entitlement and updates the beneficiary ledger in a single transaction.

Grievance & Audit Chaincode

Timestamps and records every beneficiary complaint; triggers automated escalation rules; produces an immutable audit trail accessible to CAG.

08 | Regulatory Framework

Food Security Legislation

- National Food Security Act, 2013 — eligibility and entitlement
- Essential Commodities Act, 1955 — procurement & storage norms
- PMGKAY (PM Garib Kalyan Anna Yojana) scheme compliance
- ONORC portability mandate — One Nation One Ration Card rules

Digital Identity & Data Law

- Aadhaar Act, 2016 — legal basis for biometric authentication
- Digital Personal Data Protection Act, 2023 (DPDP Act)
- IT Act, 2000 — electronic records & digital signatures
- UIDAI Aadhaar Authentication Usage Policy

Finance & Procurement

- PFMS (Public Financial Management System) API alignment
- DBT Mission framework — Direct Benefit Transfer norms
- CVC Guidelines on procurement transparency
- CAG audit compliance — immutable ledger as audit evidence

Blockchain & Technology

- MeitY National Blockchain Framework, 2021
- NITI Aayog Blockchain Strategy (PDS use case cited)
- ISO/TC 307 DLT Standards for permissioned systems
- NIC Security Policy for Government Cloud Applications

09 | Security, Privacy, Scalability & Performance

Security

- x.509 Certificate-based MSP identity
- HSM-backed private key storage
- TLS 1.3 on all peer communications
- CERT-In compliant tamper-proof audit log
- Multi-signature transaction endorsement

Privacy

- Private Data Collections (PDC) per channel
- Aadhaar biometrics never stored on-chain
- Role-based data isolation per channel
- DPDP Act 2023 data minimization
- ZK proof for eligibility verification

Scalability

- Channel sharding per state & district
- 8,000+ TPS via Raft ordering cluster
- Kubernetes horizontal peer scaling
- CouchDB rich queries on world state
- IPFS for large document storage

Performance

- <1 s transaction finality (Raft)
- 99.9% uptime — multi-AZ cloud deploy
- Offline FPS dApp with delta-sync
- CDN-cached endorsement policies
- Tested: 50,000 concurrent FPS nodes

10 | Execution Plan

Phase 1

M 1–2

Foundation

- Hyperledger Fabric network design
- Chaincode development (Go)
- NFSA & ONORC regulatory mapping
- MeitY / NIC architecture review

Phase 2

M 3–4

Integration

- Aadhaar oracle & PFMS API build
- IoT-GPS device onboarding
- FPS mobile dApp development
- Security & penetration testing

Phase 3

M 5–6

Pilot

- 500 FPS across 2 pilot districts
- 10 lakh beneficiary onboarding
- UAT & load testing
- Regulatory sandbox filing

Phase 4

M 7–8

Scale

- Full rollout — 2 states
- ONORC portability activation
- AI anomaly engine live
- Replication kit for all UTs

11 | Expected Outcomes & Pilot Deployment

40%

Reduction in leakage & diversion

₹40KCr

Potential annual subsidy savings

500 FPS

Phase 3 pilot scope

10 Lakh

Beneficiaries in pilot phase

Pilot Deployment Plan — 2 Districts, 1 State

Pilot Scope

- 500 Fair Price Shops (2 pilot districts)
- 10 lakh NFSA enrolled beneficiaries
- FCI depot + 5 state warehouses
- 3 commercial banks for DBT triggers
- 50 IoT-GPS devices on grain trucks

Timeline

- Month 7 — FPS node setup & onboarding
- Month 8 — live grain transactions begin
- Month 9 — Aadhaar entitlement active
- Month 9.5 — AI anomaly engine active
- Month 10 — regulatory review & report

Eval. Metrics

- Leakage reduction target: $\geq 40\%$
- Beneficiary satisfaction NPS ≥ 55
- Transaction finality < 1 second
- System uptime SLA: 99.9%
- Grievance resolution time < 3 days

12 | Government Collaboration & Business Model

Government Partners

DFPD / MoCAFPD

FCI / NAFED

State Food Depts.

MeitY / NIC

UIDAI (Aadhaar)

DBT Mission / PFMS

Replicable across all 36 States & UTs via standardised node deployment kits.

Business Model & Proliferation

Govt SaaS Licensing

Annual per-state subscription tiered by FPS count; concessional pricing for NE states and UTs.

Central Scheme Co-Funding

Eligible under PFMS-linked Digital PDS modernisation; NIC / MeitY grant pathway available.

Device Lease Revenue

IoT-GPS device lease + maintenance per vehicle; dApp licensing per registered FPS dealer.

Open API Marketplace

Third-party agri-fintech and NGO access via tiered API subscriptions on call-volume basis.

13 | Additional Details

Market Opportunity

- ₹2.05 lakh crore annual food subsidy — India's largest welfare programme
- 5.4 lakh Fair Price Shops — all candidates for digital node onboarding
- 81 crore NFSA beneficiaries across 36 states and UTs
- GovTech PDS modernisation market growing at ~18% CAGR through 2028

User Adoption Strategy

- FPS dealer training via Krishi Vigyan Kendra & CSC network
- Offline-first dApp works reliably on 2G/3G feature phones
- Beneficiary awareness via IVRS, SMS alerts and Gram Panchayat camps
- Dedicated multilingual toll-free helpline for FPS and beneficiaries

Strategic Partnerships

- NIC — government cloud hosting and CERT-In security review
- UIDAI — Aadhaar Authentication API and biometric SDK licensing
- ISRO NAVIC — GPS fallback for remote and hilly districts
- State IT Missions — ground-level implementation and support

Risk Mitigation

- Biometric failure → OTP-based fallback with immutable audit log
- Connectivity outage → offline dApp with encrypted delta-sync on resume
- Key compromise → HSM + multi-sig + automated key-rotation policy
- State resistance → phased onboarding with performance-linked incentives

Founding Team

Fo

Founder & CEO

[Founder Name]

[Blockchain / GovTech background]

CT

CTO

[CTO Name]

[Hyperledger Fabric / Distributed Systems]

Pr

Product Lead

[Product Name]

[PDS Domain / Government IT]

AI

AI / ML Lead

[ML Lead Name]

[Anomaly Detection / Analytics]

Signatures

Founder & CEO

CTO

Product Lead

AI / ML Lead



GrainChain India

Blockchain-Enabled Trust for India's Public Distribution System

Immutable

Transparent

Automated

Interoperable

Compliant